

FinSight | Software Test Plan

Extracted from the FinSight Final Year Project report, Chapter 7. Original text and table entries are retained; layout has been reformatted for portrait reading.

CHAPTER 7: Testing Plan

7.1 Test Objectives

The primary objective is to validate that the FinSight application functions according to the Software Requirements Specification (SRS) and meets the acceptance criteria set by the stakeholders.

7.1.2 Scope

Scope: All functional modules listed in the System Description, including Register, Login (2FA), Manage Profile, Income Module, and Admin User Management .

Out of Scope: Performance testing (load/stress testing beyond basic input handling), third-party bank API integrations (simulated only if necessary) and hardware-specific testing (biometrics simulation only).

7.1.3 Scope

The following specific features will be tested based on the Use Cases :

User Features: Profile Editing, 2FA Setup, Password Verification, Income Management (Add/Edit/View/Delete), Export Reports.

Admin Features: Modify User Profiles, Reset Passwords, Disable/Suspend Users.

7.1.3.1 Feature Tested

There is the concept of risk-based testing. It entails the process of carrying out a risk analysis under the topic of Risk-Based Testing in the Risk Analysis 3.9.2.1. The outcome of the risk analysis helped to identify the type of features to be tested. The prioritization is guided by the likelihood of failing, impact on the system, overall risk, and the level of priority. A high-risk attribute will be tested first to make critical functionalities more stable.

7.1.4 Test Process

The testing process follows these stages:

Planning: Defining the test strategy and scope (STP).

Analysis & Design: Creating Test Cases based on Use Cases (Section 4.0).

Execution: Running the test cases manually on the test environment.

Reporting: Logging results and defects.

Closure: confirming exit criteria are met and archiving testware.

7.1.5 Item Pass/Fail Criteria

Pass: The actual result matches the expected result defined in the test case

specification.

Fail: The actual result deviates from the expected result, or a system crash occurs.

7.1.6 Suspension Criteria and Resumption Requirements

7.1.6.1 Suspension Criteria

The following are conditions under which testing activities for the FinSight system are immediately suspended:

System not available: In case the FinSight web application becomes unavailable or crashes very often and a test case execution is not possible.

Critical Defects: If a critical defect is found that prevents dependent modules from being tested, such as a "showstopper" in the Login or Income Management module.

Environment Failure: If in the test environment major physical or technical disturbances took place, which include server crashes, database corruption, or any power failure that makes the test environment unstable.

7.1.6.2 Suspension Criteria

In case the test is suspended on account of any of the above factors, it will recommence when the following conditions are satisfied:

System Availability: FinSight application is available.

Defect Resolution: The key defect, which has caused the blockage, has been resolved by the development team.

Environment Stability: The environment for the test, which involves the servers, database, and the environment itself, has been completely rejuvenated and confirmed to be stable before moving ahead with testing.

7.1.7 Entry and Exit Criteria

This section defines what must be present to begin testing officially-the Entrance Criteria-and the conditions needed to bring about the proper completion of this phase of testing, the Exit Criteria. Defining the criteria ensures that testing is performed efficiently and cannot be open-ended.

7.1.7.1 Entrance Criteria

Before embarking on the Execution phase of testing, the following requirements need to be met:

Resources Available: The testing equipment and/or hardware and software (sample testing equipment, browsers) are all accessible and ready for use.

Environment Ready: The environment has now been successfully established and tested for its readiness.

Documentation Approved: Software Requirements Specifications

Documents SRS documents are approved by the concerned project stakeholders, for example, SRS documents are Project Supervisor approved documents.

Pre-Test Completed: The previous levels of testing like Unit Testing and Integration Testing have already been performed by the developers, which came out satisfactory.

7.1.8 Exit Criteria

The testing activities will be considered complete when the following criteria are met:

Test Coverage Achieved: All specified test cases have been executed, achieving a test coverage of at least 80%. This is calculated using the formula:

Figure Test Coverage Formula

Critical Defects Fixed: All defects with a priority level of High or Critical (e.g., data loss in Income Management or security breaches in 2FA) have been resolved and closed.

Defect Documentation: Any remaining unresolved defects (Low/Medium priority) are documented in the Test Summary Report with details on their severity and impact.

Timeline Met: The software testing phase is completed within the planned project timeline.

7.1.9 Risks and Contingencies

There are some issues related to risk that need to be planned for during testing, as they might become a cause for some substantial problems regarding the development and testing schedules of the product. Although product risks are defined within the risk-based testing document, these are the identified project risks for FinSight:

Table Risks and Contingencies

ID No	Project Risks	Contingencies	Impact
1	Unavailability of FinSight Web Application	To compensate for the inability of obtaining time, increase personnel/effort. There could also be a need to minimize test cases of lowest priority.	Unable to run the test and causes a delay in the project.
2	Unavailability of Test Environment	Planning and preparing a checklist for ensuring requirements in the test environment (Server/Database) are in place. In the event of an outage, work on documentation or offline activities.	Unable to run tests and results in delay.
3	Lack of Knowledge Resources	Seek advice from the supervisor or subject matter experts to get a better understanding of the intricate financial calculations. Look for training or documentation.	Test the fallibility or untimeliness of validating complex logic.
4	Time and Cost Limitation	Reduce or avoid low-risk/priority test cases (as identified by Risk Analysis) to allow higher priority ones like financial modules to be tested.	Insufficient time to execute test cases; possibility of coverage holes.

ID No	Project Risks	Contingencies	Impact
5	Large Number of Defects Found	Critical bugs (Calculation errors, Login fails) should first be addressed. Programmers and testers must collaborate in order to handle "showstoppers" ASAP.	Incomplete at performing dependent test cases; testing bottleneck.
6	Lack of Test Data	Conduct testing by using the generation of synthetic dummy data. Prepare comprehensive datasets of income and expenses prior.	Test delays and the risk of inaccuracies while validating financial plans.
7	Changes of Original Requirements	Apply Agile flexibility principles. If changes to requirements occur (such as inclusion of a new source of income), postpone non-critical tests to fit new scope into the testing efforts.	Defer the running of the test; requires an update on the test cases.

7.1.10 Test Level

The FinSight Financial Insight system will, therefore, be integrated with different levels of testing to ensure that every functionality serves its purpose as required. The major test level to be applied on this project is Acceptance Testing.

Acceptance Testing At this level, the system is tested to verify whether it satisfies all specified requirements and behaves accordingly as expected by an end-user. This testing will ensure that the application is actually ready to be used in the real world. To perform this, black-box testing will be carried out, where only the functionality of the system will be tested without looking at the internal code. The tests will be based on the SRS document. The tests will be performed by regular users and admins with the help of developers, which will validate the usability of the system in a controlled environment.

7.1.10.1 Black Box Testing

In the black-box testing approach, only the inputs and outputs of the FinSight system are to be considered in order to validate its interactions. This approach is suitable for acceptance testing since functionalities like User Authentication, Income Management, and 2FA Verification are checked without looking into the internal code structure and logic. In this particular project, this approach is considered as Specification-Based Testing, where all test cases and validations should strictly adhere to the specifications mentioned in the SRS document.

7.1.10.2 Risks Based Testing

A risk-based testing strategy will also be adopted in order to effectively prioritize tests. Given the time constraint, as well as the fact that financial data processing in FinSight is a critical activity, the testing strategy in this assignment has adopted the approach of testing critical areas first.

It entails evaluating the risk risk by determining the probability of failure and the consequent risk. The process of performing risk risk includes the following steps:

Analyze requirements and the SRS document.

Identifying the risk can be achieved by determining the "Likelihood" and "Impact" for each failure of the requirements. Generally, this calculation can be carried out in terms of potential costs

Use Total Risk to assess priorities of risk by multiplying Likelihood and Impact.

7.2 UC-Test Case Summary

Table UC-Test Case Summary

Use Case ID	Use Case Name	Test Case ID	Test Case Name
UC_01	Edit Personal Info	TC01.1	Functionality Test Case for Edit Personal Info
UC_01	Edit Personal Info	TC01.2	Validation and Error Handling for Edit Personal Info

Use Case ID	Use Case Name	Test Case ID	Test Case Name
UC_01	Edit Personal Info	TC01.3	Catastrophic Test Case for Edit Personal Info
UC_04	Add New Income	TC04.1	Verify Successful Addition for Add New Income
UC_04	Add New Income	TC04.2	Verify System Rejection for Add New Income
UC_04	Add New Income	TC04.3	Verify System Stability with Excessive Data Input for Add New Income
UC_06	View History	TC06.1	Verify Successful Retrieval and Filtering of Income History for View History
UC_06	View History	TC06.2	Verify System Behavior When No Income History Exists for View History
UC_07	Authenticate Credentials	TC07.1	Functionality Test Case for Authenticate Credentials
UC_07	Authenticate Credentials	TC07.2	Validation and Error Handling Test Case for Authenticate Credentials
UC_07	Authenticate Credentials	TC07.3	Catastrophic Test Case for Authenticate Credentials
UC_02	Register Account	TC02.1	Successful Registration
UC_02	Register Account	TC02.2	Invalid Email Format
UC_02	Register Account	TC02.3	Duplicate Email
UC_03	Add New Expense	TC03.1	Successful Addition
UC_03	Add New Expense	TC03.2	Missing Required Fields
UC_03	Add New Expense	TC03.3	Invalid Amount Format
UC_05	OCR Receipt Scan	TC05.1	Successful Scan & Detection
UC_05	OCR Receipt Scan	TC05.2	Unreadable Image Input
UC_05	OCR Receipt Scan	TC05.3	Save as Expense or Income
UC_08	Tax Calculation	TC08.1	Successful Tax Estimation
UC_08	Tax Calculation	TC08.2	Zero Income Scenario
UC_09	View Financial Charts	TC09.1	Charts Display Correctly
UC_09	View Financial Charts	TC09.2	Empty Data Scenario